ST MARY'S DSG, KLOOF

GRADE: 10

8 September 2023

PHYSICAL SCIENCE CYCLE TEST
KINEMATICS

TIME: 55 MINS (incl. reading time)
ASSESSOR: T. Kirsten

TOTAL: 63
MODERATORS: K. Bolton and G. Main

NAME: MEMO

SCIENCE TEACHER: _____

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 7 pages. A data sheet is provided.
2. Answer on the question paper in the spaces.
3. Write your name and your science teacher's name on your question paper.

QUESTION 1 – MULTIPLE CHOICE - Choose the letter that you consider to be most correct and write your choice in the block alongside the question.

1.1 A student wanted to calculate the final velocity of a toy car, starting from rest, accelerating at a rate of 0,2 $m.s^{-2}$, and moving for 10 s. The formula the student used was:

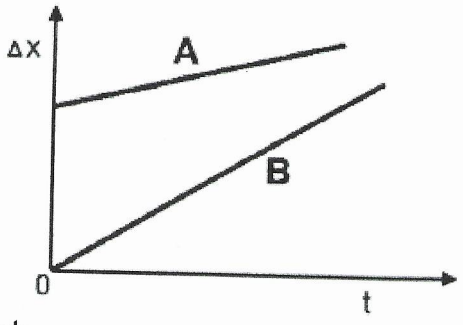
$$v_f = v_i + a\Delta t$$

How many vector and scalar quantities does this formula contain?

	VECTOR QUANTITIES	SCALAR QUANTITIES
A	2	2
B	3	1
C	1	3
D	4	0

B

1.2 The motion of two objects, A and B, is represented in the following displacement-time graph.



The graph indicates that:

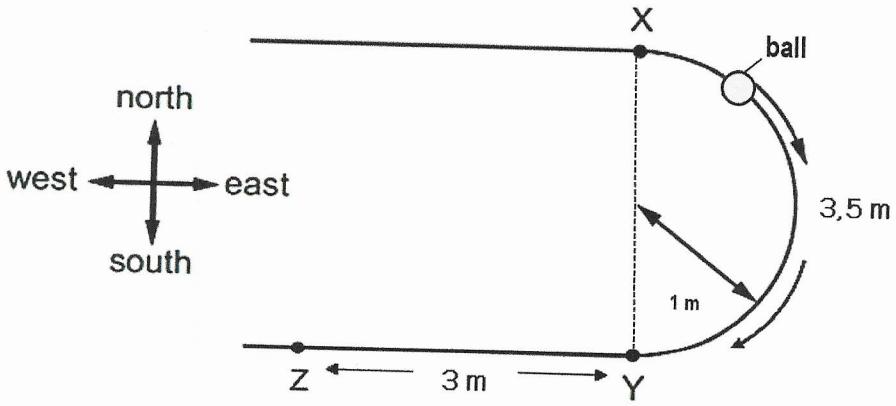
A B is moving faster than A.
B A and B have different accelerations.
C A and B are on a collision course.
D A and B are moving in opposite directions.

A

2 x 2 = [4]

QUESTION 2

A ball travels from Point X to Point Y, a distance of 3,5 m, at a CONSTANT SPEED along a horizontal semi-circular track of **radius** 1 m in 2 s, as shown below.



2.1 What is the displacement of the ball from X to Y? (2)

2m ✓ South ✓

2.2 Define *velocity*. (2)

Rate of change ✓ of displacement ✓

2.3 Calculate the average velocity of the ball between X and Y. (3)

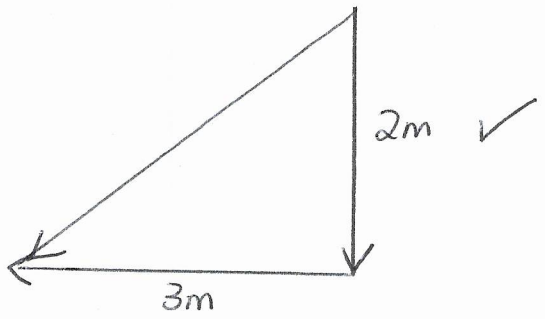
$$\text{Av. velocity} = \frac{\text{total } \Delta x}{\text{total } t} = \frac{2m}{2s} \checkmark$$

$$= 1 m \cdot s^{-1} \checkmark \text{ South} \checkmark$$

2.4 How would the average speed of the ball between X and Y differ to your answer to Q2.3? Write only MORE THAN, LESS THAN or EQUAL. (1)

More ✓

2.5 From Point Y, the ball then travels to Point Z, a distance of 3 m. Calculate the magnitude of the resultant displacement of the ball from the starting point (X). A sketch vector diagram must be included in your answer. (3)



$\vec{R} = \sqrt{2^2 + 3^2}$ ✓

$= 3,61m$ ✓

[11]

QUESTION 3

3.1 A cyclist, heading **East** at a constant velocity of 9 $m.s^{-1}$, approaches the beach on a straight tar road.

- Soon the road changes into a soft gravel road. She enters the soft gravel road, and slows down to a velocity of 3 $m.s^{-1}$.
- This takes place over a time of 4 s on the soft gravel road.

3.1.1 Define *acceleration*. (2)

Rate of change of velocity ✓✓

3.1.2 Calculate the acceleration of the cyclist on the soft gravel road. (4)

$V_i = 9$
$V_f = 3$
$t = 4s$
$a = ?$

$V_f = V_i + a\Delta t$ ✓

$3 = 9 + a \cdot 4$ ✓

$\frac{3-9}{4} = a$

$\therefore a = -1,5 m.s^{-2}$

ie $1,5 m.s^{-2}$ ✓ West ✓

3.2 The cyclist, travelling at the new velocity of 3 m.s^{-1} on the gravel road, sees a rock in the road 5 m ahead. She applies her brakes and slows down at a rate of 1 m.s^{-2} before stopping. Show by calculation if the cyclist will hit the rock or not. (4)

$V_i = 3$
$V_f = 0$
$\Delta x = ?$
$a = -1$

$V_f^2 = V_i^2 + 2a\Delta x$ ✓
$0 = (3)^2 + 2(-1)\Delta x$ ✓
$-9 = -2\Delta x$
$\therefore \Delta x = 4,5m$ ✓

No ✓ she will not hit the rock (4,5m < 5m)

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QUESTION 4

The table below shows the minimum stopping distances required for a car travelling at various speeds on a dry road with good brakes and good tyres. The car is travelling **North**.

Initial speed of car (km.h^{-1})	Distance travelled during reaction time (m)	Distance travelled while braking (m)
32	6	6
48	9	14
64	12	24
80	15	38
97	16	58

4.1 Describe the relationship between minimum stopping distance for a car and the initial speed of the car. (2)

The greater the initial speed of the car ✓, the greater the minimum stopping distance required ✓

4.2 Convert the speed of 97 km.h^{-1} into m.s^{-1}. (1)

26,94 m·s^{-1} ✓

4.3 Calculate the reaction time of the driver in the car travelling at **97 km.h^{-1}**. (3)

$V = 26,94$ (c.o.e)
$\Delta x = 16m$
$t = ?$

$\Delta x = V_i\Delta t + \frac{1}{2}a\Delta t^2$ ✓
$16 = 26,94 \cdot \Delta t + 0$ ✓
$\therefore t = 0,59s$ ✓

4.4 List 2 factors which could increase the reaction time of the driver. (2)

Being intoxicated / influence of drugs ✓

Being on cell phone ✓

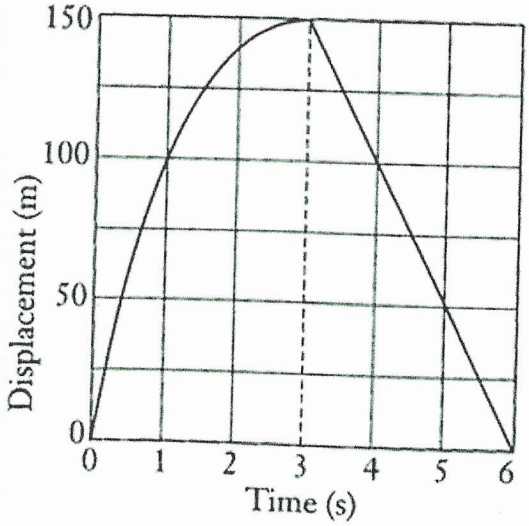
4.5 What is the distance travelled during the driver's reaction time known as? (1)

thinking distance ✓

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QUESTION 5

The displacement-time graph below represents the motion of a motorbike along a straight race track, **West**.



5.1 Describe the motion of the motor bike:

i) from 0 - 3 s. (2)

decreasing velocity ✓ West ✓

ii) from 3 - 6 s. (2)

constant velocity ✓ East ✓

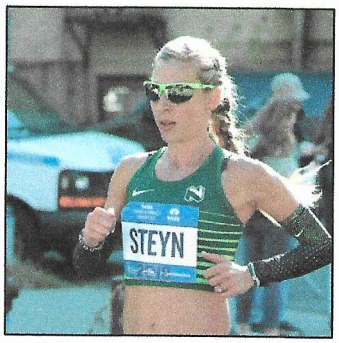
5.2 What is the displacement of the motor bike from the starting point after 6 s? (1)

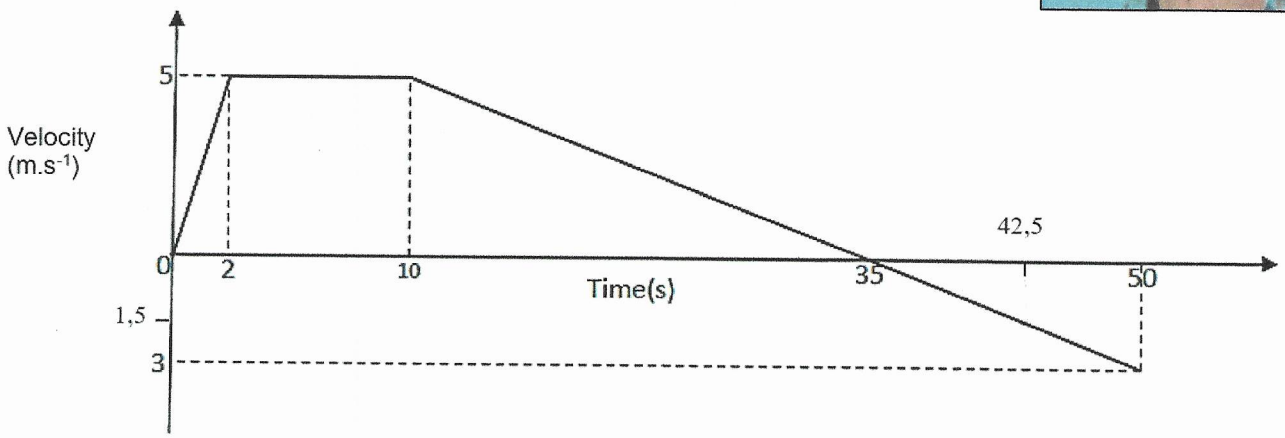
0m / Zero ✓

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QUESTION 6

Gerda Steyn is a South African marathon and ultramarathon athlete. Gerda sets out on a training run. She starts off by running in a **Southerly** direction. Her motion during the first part of her run is graphed below.





6.1 Using the graph, calculate Gerda's acceleration during the first two seconds of the run. (4)

a = gradient = $\frac{\Delta v}{\Delta t}$ ✓ = $\frac{5-0}{2-0}$ ✓

= 2,5 $m \cdot s^{-2}$ ✓ South ✓

6.2 Calculate Gerda's average velocity in the first 10 s of her run. (4)

Distance = area under graph

0–2s : $\frac{1}{2}$ 2×5 = 5m

2–10s : 8×5 = 40m

Av. velocity = $\frac{\text{total } \Delta x}{\text{total } t}$ = $\frac{45m}{10s}$ ✓ ✓

= 4,5 $m \cdot s^{-1}$ ✓ South ✓

6.3 Describe Gerda's motion between 10 and 50 s. (6)

10–35s: decreasing velocity ✓ South ✓ (uniform negative acceleration ✓) 35–50s: increasing velocity ✓ North ✓ (uniform positive acceleration ✓)

6.4 What is Gerda's instantaneous velocity at 42,5 s? (2)

1,5 $m \cdot s^{-1}$ ✓ North ✓

6.5 Calculate Gerda's displacement between 10 and 50 s (4)

Displacement = area under graph

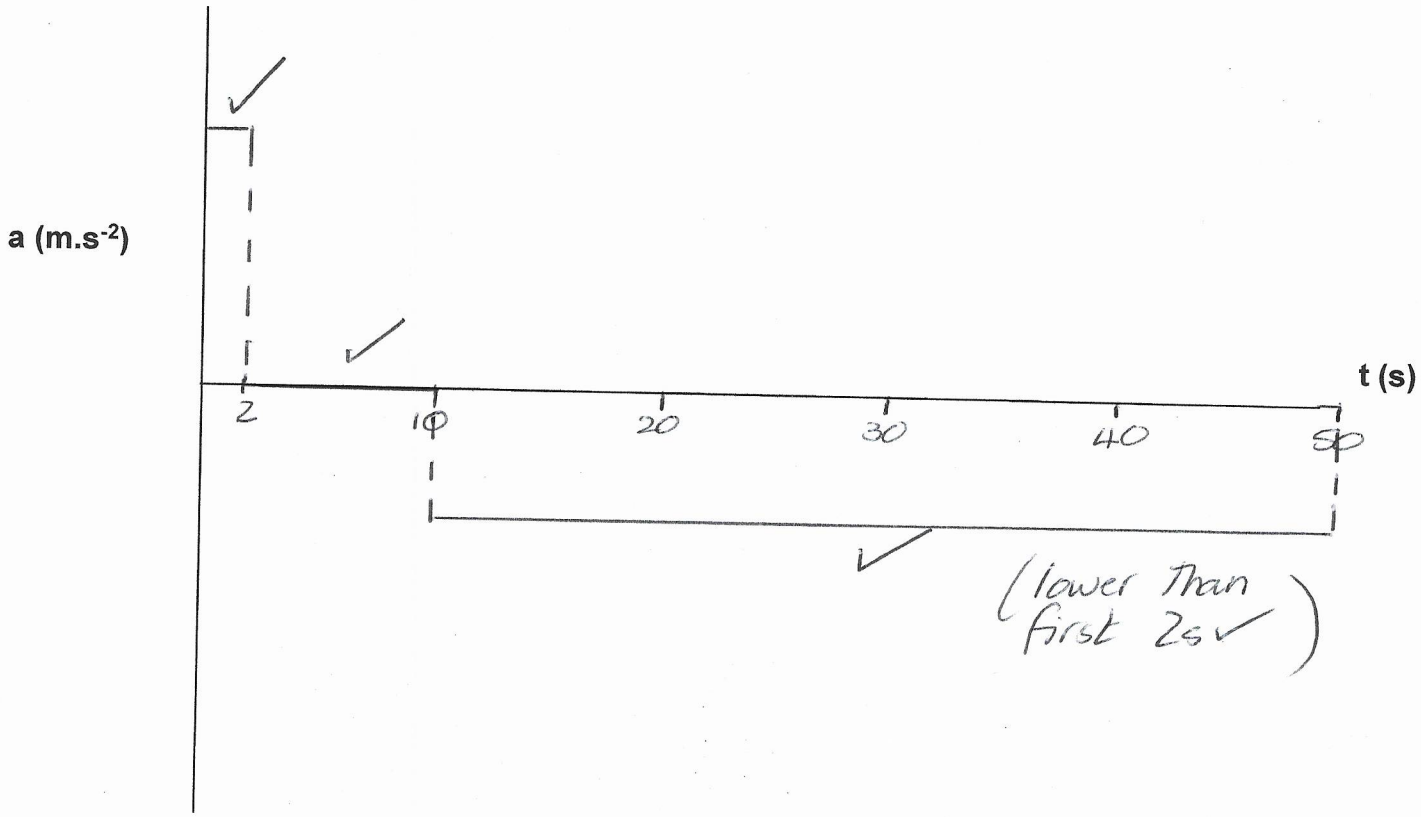
10-35s : $\frac{1}{2}$ 25 x 5 = +62,5m } ✓

35-50s : $\frac{1}{2}$ 15 x -3 = -22,5m }

Δx = 62,5 − 22,5 ✓

= 40m ✓ South ✓

6.6 On the axes below, sketch an acceleration-time graph that represents the first 50 s of Gerda's run. You need only show actual values on the time axis. It is the shape that is important. (4)



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